TEXORA · LECTURE SHEET

Alternative Methods of Joining

Welding (hot-air, RF, ultrasonic, laser), adhesive bonding, linking and seamless knitting: the non-thread joining methods, where each wins, and the material limits that confine them.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 6 · LEVEL 3 TERM 1

1. Why look beyond the needle?

Sewing pierces the material, consumes thread, and needs a skilled operator. For some products — seamless sportswear, medical disposables, waterproof shells, inflatables, lingerie — a pierced, thread-based seam is the wrong tool: needle holes leak, thread wicks water, stitches chafe. Alternative joining methods replace the thread structure with **fusion, adhesion or mechanical fastening**. None has displaced sewing generally (Topic 1 gives the reasons), but each owns a territory where it beats the needle outright.

2. Welding and fusing: joining thermoplastics with heat and pressure

METHOD	PRINCIPLE	USES AND LIMITS
Hot-air / hot-wedge welding	A jet of hot air or a heated wedge melts the surfaces (usually coated fabrics or films, or thermoplastic seam tape), rollers press them together.	Waterproof seam sealing of rainwear (tape over sewn seams), tents, inflatables. Needs thermoplastic content; not for pure cotton.
High-frequency (RF/dielectric) welding	Radio-frequency field heats polar polymers (PVC, PU) from within; die presses the joint.	PVC rainwear, medical bags, stationery, inflatables. Material must be RF-active — polyester and nylon respond poorly.
Ultrasonic joining	A horn vibrating at ~20–35 kHz melts synthetic material locally at the joint; the same head can cut and seal an edge in one pass.	Quilting, bra straps, disposable gowns/masks, lace edging, spot 'tacking'. Needs a high thermoplastic fraction; seam width and strength limited.
Laser welding / cutting	Focused beam fuses a narrow line or cuts with a sealed edge.	Precision technical textiles; high capital cost, narrow joints.

3. Adhesive bonding

Bonding spreads or films an adhesive across the joint. Forms: **hot-melt films and webs** (activated in a press or by a heated seam process — the basis of 'seamless' bonded hems, pocket flaps and bra moulding), **liquid adhesives** (for leather goods, shoe uppers, lamination), and **pressure-sensitive tapes**. A bonded seam can be completely flat, hole-free and stretch-matched to the fabric, which is why performance sportswear bonds hems and pocket openings. Its engineering controls are adhesive chemistry matched to the fibres, activation temperature/time/pressure, and wash durability testing — a bond that survives the showroom but not ten washes is a returns programme, not a technology.

4. Mechanical and other methods

- **Linking** — the knitter's joining: loops of one piece are hooked onto the loops of the next and chained off, stitch for stitch, on a linking machine. Fully-fashioned knitwear collars and shoulder seams; the seam stretches exactly as the fabric does.
- **Fasteners as joints** — rivets, snap fasteners, hook-and-loop, zips and buttons join at points rather than along a line; jeans rivets historically reinforced the pocket corners a bartack now serves.
- **Integral knitting (seamless)** — the joining step is designed out: circular and whole-garment knitting produce tubes and complete garments with no side seams at all.
- **Stapling and tagging** — legitimate only for temporary or non-apparel assembly; never a garment seam.

5. Choosing the joining method: a decision table

REQUIREMENT	SEW	WELD / BOND	LINK / KNIT-INTEGRAL
Natural fibres (cotton, wool)	Yes — the default	Generally no (no thermoplastic to fuse)	Linking for knitwear
Waterproof, hole-free seam	Only with seam-sealing tape applied after	Yes — native territory	No
Repairable, alterable garment	Yes	Difficult to unpick/rework	Partly
Stretch with knit fabric	With chain/overedge stitches	Yes, with stretch adhesives	Yes, perfectly matched
Lowest capital and widest material range	Yes	No	Specialised machines

The trend of Topic 11 builds on this sheet: modern factories mix methods on one garment — sewn structure, bonded hems, welded reinforcements — choosing per seam, not per factory.

Glossary

TERM	MEANING
Seam sealing	Applying thermoplastic tape over a sewn seam (usually by hot-air welding) to close needle holes against water.
Linking	Joining knitted pieces loop-to-loop so the seam has the same structure and stretch as the fabric.
Hot-melt film	A thermoplastic adhesive sheet activated by heat and pressure to bond plies without stitches.
Dielectric (RF) welding	Joining polar polymers by internal heating in a radio-frequency field under a shaped die.

Key points and summary

- Alternatives win where sewing's holes, thread or bulk are the problem: waterproofing, chafe-free stretch, disposables.
- Welding needs thermoplastic material; RF welding needs RF-active polymers (PVC, PU).
- Bonding quality is set by adhesive–fibre matching and activation control, and proven by wash testing.
- Linking and seamless knitting do not join pieces so much as continue the fabric structure across the junction.

Practice

Answer in your own words; brief model answers follow.

1. A PVC raincoat order needs waterproof seams. Which joining route do you specify, and why is a plain sewn seam unacceptable even with strong thread?
2. Why can a 100% cotton shirt not be ultrasonically welded, and what ultrasonic result would you expect on a 100% polyester bra strap?
3. State two advantages and two risks of replacing a sewn hem with a hot-melt bonded hem on a sports T-shirt.

Answer key

1. Weld the seams (RF or hot-air/wedge welding of the PVC) or sew and then seam-seal; an unsealed sewn seam leaks through every needle hole and along the thread, however strong it is.
2. Ultrasonic energy joins by melting thermoplastic; cotton has none, so there is nothing to fuse — the horn would only scorch or cut it. The polyester strap is fully thermoplastic and welds readily, cutting and sealing in one pass.
3. Advantages: a flat, hole-free, chafe-free hem that stretches with the fabric and looks clean. Risks: bond failure in repeated washing if adhesive or activation is wrong, and rework difficulty — a mis-bonded hem cannot be unpicked like stitching.
